

cwltool

Mode summaries

Failed cwltool tests (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests (slurm mode) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests (all platforms/modes union) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

FT3

Failed cwltool tests on ft3 (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on ft3 (slurm mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on ft3 (all modes union) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

MN5

Failed cwltool tests on mn5 (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on mn5 (slurm mode) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on mn5 (all modes union) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

LUMI

Failed cwltool tests on lumi (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on lumi (slurm mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on lumi (all modes union) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests (all platforms/modes union) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled

ID	Description
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

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Failed cwltool tests on ft3 (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on ft3 (slurm mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on ft3 (all modes union) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

MN5

Failed cwltool tests on mn5 (local mode) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on mn5 (slurm mode) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
networkaccess	Test networkaccess enabled
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

Failed cwltool tests on mn5 (all modes union) (3 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
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filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
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Failed cwltool tests on lumi (all modes union) (2 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut

streamflow

Mode summaries

Failed streamflow tests (local mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2

ID	Description
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests (slurm mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input

ID	Description
cl.optional_bindings_provided	cl_optional_bindings_provided
cl.optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool

ID	Description
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests (all platforms/modes union) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code

ID	Description
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker

ID	Description
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

FT3

Failed streamflow tests on ft3 (local mode) (33 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_empty_array_input	cl_empty_array_input
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
env_home_tmpdir_docker	env_home_tmpdir_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative

ID	Description
metadata	metadata
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

Failed streamflow tests on ft3 (slurm mode) (26 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
directory_input_docker	directory_input_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays

ID	Description
output_secondaryfile_optional	output_secondaryfile_optional
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

Failed streamflow tests on ft3 (all modes union) (43 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
env_home_tmpdir_docker	env_home_tmpdir_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output

ID	Description
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative_metadata	json_output_path_relative_metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

MN5

Failed streamflow tests on mn5 (local mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess

ID	Description
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests on mn5 (slurm mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path

ID	Description
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker

ID	Description
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests on mn5 (all modes union) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output

ID	Description
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

LUMI

Failed streamflow tests on lumi (local mode) (1 unique failure)

ID	Description
iwd-container-entryname1	iwd-container-entryname1

Failed streamflow tests on lumi (all modes union) (1 unique failure)

ID	Description
iwd-container-entryname1	iwd-container-entryname1

Failed streamflow tests (all platforms/modes union) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput

ID	Description
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs

ID	Description
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

FT3

Failed streamflow tests on ft3 (local mode) (33 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_empty_array_input	cl_empty_array_input
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
env_home_tmpdir_docker	env_home_tmpdir_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entrystate1	iwd-container-entrystate1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative_metadata	json_output_path_relative_metadata
no_inputs_commandlinetool	no_inputs_commandlinetool

ID	Description
no_inputs_workflow	no_inputs_workflow
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

Failed streamflow tests on ft3 (slurm mode) (26 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
directory_input_docker	directory_input_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
output_secondaryfile_optional	output_secondaryfile_optional

ID	Description
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

Failed streamflow tests on ft3 (all modes union) (43 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
env_home_tmpdir_docker	env_home_tmpdir_docker
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr

ID	Description
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative_metadata	json_output_path_relative_metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple

MN5

Failed streamflow tests on mn5 (local mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation

ID	Description
cl.empty_array_input	cl.empty_array_input
cl.optional_bindings_provided	cl.optional_bindings_provided
cl.optional_inputs_missing	cl.optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow

ID	Description
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shortcut	stdin_shortcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests on mn5 (slurm mode) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker

ID	Description
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker

ID	Description
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

Failed streamflow tests on mn5 (all modes union) (51 unique failures)

ID	Description
booleanflags_cl_noinputbinding	booleanflags_cl_noinputbinding
cl_basic_generation	cl_basic_generation
cl_empty_array_input	cl_empty_array_input
cl_optional_bindings_provided	cl_optional_bindings_provided
cl_optional_inputs_missing	cl_optional_inputs_missing
colon_in_output_path	colon_in_output_path
colon_in_paths	colon_in_paths
cwloutput_nolimit	cwloutput_nolimit
directory_input_docker	directory_input_docker
docker_json_output_location	docker_json_output_location
docker_json_output_path	docker_json_output_path
dockeroutputdir	dockeroutputdir
env_home_tmpdir_docker	env_home_tmpdir_docker
env_home_tmpdir_docker_no_return_code	env_home_tmpdir_docker_no_return_code
expr_reference_self_noinput	expr_reference_self_noinput
filesarray_secondaryfiles	filesarray_secondaryfiles
format_checking_equivalentclass	format_checking_equivalentclass
format_checking_subclass	format_checking_subclass
hints_unknown_ignored	hints_unknown_ignored
initial_workdir_empty_writable_docker	initial_workdir_empty_writable_docker
initial_workdir_output	initial_workdir_output
initial_workdir_secondary_files_expr	initial_workdir_secondary_files_expr

ID	Description
initworkdir_expreng_requirements	initworkdir_expreng_requirements
inlinejs_req_expressions	inlinejs_req_expressions
input_file_literal	input_file_literal
iwd-container-entryname1	iwd-container-entryname1
iwd-passthrough2	iwd-passthrough2
iwdr_dir_literal_real_file	iwdr_dir_literal_real_file
json_output_location_relative	json_output_location_relative
json_output_path_relative	json_output_path_relative
legal_symlink	legal_symlink
metadata	metadata
nested_cl_bindings	nested_cl_bindings
nested_prefixes_arrays	nested_prefixes_arrays
networkaccess	networkaccess
no_inputs_commandlinetool	no_inputs_commandlinetool
no_inputs_workflow	no_inputs_workflow
no_outputs_commandlinetool	no_outputs_commandlinetool
no_outputs_workflow	no_outputs_workflow
output_secondaryfile_optional	output_secondaryfile_optional
packed_import_schema	packed_import_schema
record_order_with_input_bindings	record_order_with_input_bindings
stdin_shorcut	stdin_shorcut
stdinout_redirect_docker	stdinout_redirect_docker
stdout_redirect_docker	stdout_redirect_docker
stdout_redirect_mediumcut_docker	stdout_redirect_mediumcut_docker
stdout_redirect_shortcut_docker	stdout_redirect_shortcut_docker
valuefrom_constant_overrides_inputs	valuefrom_constant_overrides_inputs
wf_compound_doc	wf_compound_doc
wf_simple	wf_simple
writable_stagedfiles	writable_stagedfiles

LUMI

Failed streamflow tests on lumi (local mode) (1 unique failure)

ID	Description
iwd-container-entryname1	iwd-container-entryname1

Failed streamflow tests on lumi (all modes union) (1 unique failure)

ID	Description
iwd-container-entryname1	iwd-container-entryname1

toil

Mode summaries

Failed toil tests (local mode) (9 unique failures)

ID	Description
exprtool_file_literal	Test file literal output created by ExpressionTool
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
scatter_embedded_subworkf low	Test simple scatter over an embedded subworkflow
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crosspr oduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_cross product	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_c rossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_neste d_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests (slurm mode) (50 unique failures)

ID	Description
FAIL_NAME	symlink to file inside of working directory should be retrieved
booleanflags_cl_noinputbin ding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line

ID	Description
cl.optional_bindings_provided	Test command line with optional input (provided)
cl.optional_inputs_missing	Test command line with optional input (missing)
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
directory_input_docker	Test directory input in Docker
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_unknown_ignored	Test unknown hints are ignored.
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool

ID	Description
input_file_literal	Test file literal as input
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
metadata	Test metadata
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
packed_import_schema	SchemaDefRequirement with \$import, and packed
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
wf_compound_doc	Test compound workflow document
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input

ID	Description
wf_scatter_twoparam_nest d_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
writable_stagedfiles	Test writable staged files.

Failed toil tests (batch mode) (313 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line

ID	Description
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nons cattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nons cattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nons cattered_true	Simple scatter: conditional on a non scattered variable (True)
condifional_scatter_on_nons cattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_ nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_sc atter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_sc atter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fa il	Non-boolean values from "when" should fail
conditionals_non_boolean_fa il_nojs	Non-boolean values from "when" should fail; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_ expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_ static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_path_notfound_war ning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_res ult	Simplest conditional pattern (True)
direct_optional_nonnull_res ult_nojs	simplest conditional pattern (false), no javascript

ID	Description
direct_optional_null_result	Simplest conditional pattern (False)
direct_optional_null_result_nojs	simplest conditional pattern (true), no javascript
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.

ID	Description
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null	pickValue: first_non_null second item is non null
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.

ID	Description
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_recurse_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../

ID	Description
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job.input.secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job.input.subdir.primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'

ID	Description
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs

ID	Description
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval_exitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.

ID	Description
record_output_file_entry_for_mat	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.

ID	Description
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step

ID	Description
step_input_default_value_override_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_override_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_override_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_override_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_from_expression_wf	Test expression in time limit in workflow
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.

ID	Description
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method

ID	Description
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow

ID	Description
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

Failed toil tests (all platforms/modes union) (314 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list

ID	Description
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files

ID	Description
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nonscattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nonscattered_true	Simple scatter: conditional on a non scattered variable (True)
condifional_scatter_on_nonscattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fail	Non-boolean values from "when" should fail
conditionals_non_boolean_fail_nojs	Non-boolean values from "when" should fail; no javascript

ID	Description
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result	Simplest conditional pattern (True)
direct_optional_nonnull_result_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result	Simplest conditional pattern (False)
direct_optional_null_result_nojs	simplest conditional pattern (true), no javascript
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.

ID	Description
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
docker_output_dir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output

ID	Description
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
exprtool_file_literal	Test file literal output created by ExpressionTool
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null	pickValue: first_non_null second item is non null
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline

ID	Description
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'

ID	Description
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval.ExitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories

ID	Description
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority

ID	Description
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax

ID	Description
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal _with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut _docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_noexp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no source	Test use default value on step input parameter with empty source
step_input_default_value_nu llsource	Test use default value on step input parameter with null source
step_input_default_value_ov erriden	Test default value on step input parameter overridden by provided source
step_input_default_value_ov erriden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_ov erriden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_ov erriden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_ov erriden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_ov erriden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up

ID	Description
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_from_expression_wf	Test expression in time limit in workflow
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)

ID	Description
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow

ID	Description
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after

ID	Description
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

FT3

Failed toil tests on ft3 (local mode) (5 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input

Failed toil tests on ft3 (slurm mode) (7 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
scatter_embedded_subworkf low	Test simple scatter over an embedded subworkflow
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crosspr oduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_cross product	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_c rossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_neste d_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on ft3 (batch mode) (219 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph.no_ default	Test use of \$graph without specifying which process to run
booleanflags.cl.noinputbin ding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl.basic_generation	General test of command line generation
cl.empty_array_input	Test that empty array input does not add anything to command line

ID	Description
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.

ID	Description
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nestaddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline

ID	Description
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwd_r_dir_literal_real_file	Test directory literal containing a real file
job.input.secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job.input.subdir.primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
listing_outputBinding.loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings

ID	Description
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
quoting_multiple_backslashes	Test quoting multiple backslashes
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow

ID	Description
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool

ID	Description
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nulsource	Test use default value on step input parameter with null source
step_input_default_value_override	Test default value on step input parameter overridden by provided source
step_input_default_value_override_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_override_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_override_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_override_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_override_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
success_codes	Test successCodes
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
valuefrom_constant_override_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool

ID	Description
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valuefrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valuefrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valuefrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior

ID	Description
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

Failed toil tests on ft3 (all modes union) (220 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement

ID	Description
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.

ID	Description
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles

ID	Description
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input

ID	Description
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions

ID	Description
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.

ID	Description
legal_symlink	symlink to file inside of working directory should be retrieved
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
quoting_multiple_backslashes	Test quoting multiple backslashes
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)

ID	Description
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal

ID	Description
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut _docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
success_codes	Test successCodes
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs

ID	Description
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input

ID	Description
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after

ID	Description
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

MN5

Failed toil tests on mn5 (local mode) (4 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input

Failed toil tests on mn5 (slurm mode) (45 unique failures)

ID	Description
FAIL_NAME	symlink to file inside of working directory should be retrieved

ID	Description
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
directory_input_docker	Test directory input in Docker
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_unknown_ignored	Test unknown hints are ignored.
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initworkdir_expr_engine_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_file_literal	Test file literal as input
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
metadata	Test metadata
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
packed_import_schema	SchemaDefRequirement with \$import, and packed
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
wf_compound_doc	Test compound workflow document
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_simple	Test simple workflow

ID	Description
writable_stagedfiles	Test writable staged files.

Failed toil tests on mn5 (batch mode) (226 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
booleanflags.cl.noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl.basic_generation	General test of command line generation
cl.empty_array_input	Test that empty array input does not add anything to command line
cl.gen_arrayofarrays	Test command line generation of array-of-arrays
cl.optional_bindings_provided	Test command line with optional input (provided)
cl.optional_inputs_missing	Test command line with optional input (missing)
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.

ID	Description
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nonscattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fail	Non-boolean values from "when" should fail
conditionals_non_boolean_fail_nojs	Non-boolean values from "when" should fail; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores

ID	Description
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result_nojs	simplest conditional pattern (true), no javascript
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript

ID	Description
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)

ID	Description
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.

ID	Description
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript

ID	Description
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_missing	Test checking when secondaryFiles are missing
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename

ID	Description
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step

ID	Description
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_from_expression_wf	Test expression in time limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow

ID	Description
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_parseInt	Test two step workflow with imported tools
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

Failed toil tests on mn5 (all modes union) (228 unique failures)

ID	Description
FAIL_NAME	symlink to file inside of working directory should be retrieved
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter

ID	Description
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nons cattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nons cattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_ nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_sc atter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_sc atter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fa il	Non-boolean values from "when" should fail
conditionals_non_boolean_fa il_nojs	Non-boolean values from "when" should fail; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_ expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_ static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_res ult_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result_ nojs	simplest conditional pattern (true), no javascript
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal _file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal

ID	Description
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement

ID	Description
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nestaddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json

ID	Description
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default

ID	Description
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.

ID	Description
scatter_embedded_subworkf low	Test simple scatter over an embedded subworkflow
scatter_multi_input_embed ded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditi onal_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enu m_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_re cords	Test secondaryFiles on output record fields
secondary_files_missing	Test checking when secondaryFiles are missing
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_ overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal _with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection

ID	Description
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_from_expression_wf	Test expression in time limit in workflow
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_wf_step	Test valueFrom on workflow step.

ID	Description
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_parseInt	Test two step workflow with imported tools
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow

ID	Description
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

LUMI

Failed toil tests on lumi (local mode) (7 unique failures)

ID	Description
exprtool_file_literal	Test file literal output created by ExpressionTool
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on lumi (slurm mode) (5 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark

ID	Description
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on lumi (batch mode) (268 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation

ID	Description
cl.empty_array_input	Test that empty array input does not add anything to command line
cl.gen_arrayofarrays	Test command line generation of array-of-arrays
cl.optional_bindings_provided	Test command line with optional input (provided)
cl.optional_inputs_missing	Test command line with optional input (missing)
clt.any_input_with_file_provided	Test Any parameter with file input to a tool
clt.any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt.any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt.any_input_with_record_provided	Test Any parameter with record input to a tool
clt.any_input_with_string_provided	Test Any parameter with string input to a tool
clt.file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt.file_size_property_with_multi_file	Test use of size in expressions for a few files
clt.optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt.optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt.optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt.optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nons cattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nons cattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nons cattered_true	Simple scatter: conditional on a non scattered variable (True)
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_ nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_sc atter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_sc atter_nojs	nested cross product scatter with condition on one dimension; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_ expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_ static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_war ning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_res ult	Simplest conditional pattern (True)
direct_optional_null_result	Simplest conditional pattern (False)
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal _file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output

ID	Description
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrpoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them

ID	Description
expr.reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_second_non_null	pickValue: first_non_null second item is non null
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions

ID	Description
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.

ID	Description
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval.ExitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories

ID	Description
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_for_mat	Test format on output record fields
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow_low	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow

ID	Description
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def-anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection

ID	Description
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs

ID	Description
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method

ID	Description
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test when step source is a single-item list and there is no linkMerge, then it not wrapped in a list, and that MultipleInputFeatureRequirement is not required.
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior

ID	Description
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

Failed toil tests on lumi (all modes union) (270 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool

ID	Description
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nonscattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nonscattered_true	Simple scatter: conditional on a non scattered variable (True)
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly

ID	Description
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result	Simplest conditional pattern (True)
direct_optional_null_result	Simplest conditional pattern (False)
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
exprtool_file_literal	Test file literal output created by ExpressionTool
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.

ID	Description
first_non_null_second_non_null	pickValue: first_non_null second item is non null
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_recur_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline

ID	Description
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions

ID	Description
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexpr	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval_exitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false

ID	Description
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_for_mat	Test format on output record fields
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields

ID	Description
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source

ID	Description
step_input_default_value_override_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_override_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_override_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_override_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_override_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs

ID	Description
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list

ID	Description
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test when step source is a single-item list and there is no linkMerge, then it not wrapped in a list, and that MultipleInputFeatureRequirement is not required.
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow

ID	Description
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

Failed toil tests (all platforms/modes union) (314 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript

ID	Description
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_n ojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_ default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_ default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbin ding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provid ed	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_prov ided	Test Any parameter with file input to a tool
clt_any_input_with_integer_ provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_a rray_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_p rovided	Test Any parameter with record input to a tool
clt_any_input_with_string_p rovided	Test Any parameter with string input to a tool
clt_file_size_property_with_e mpty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_ multi_file	Test use of size in expressions for a few files

ID	Description
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nonscattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nonscattered_true	Simple scatter: conditional on a non scattered variable (True)
condifional_scatter_on_nonscattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fail	Non-boolean values from "when" should fail
conditionals_non_boolean_fail_nojs	Non-boolean values from "when" should fail; no javascript

ID	Description
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result	Simplest conditional pattern (True)
direct_optional_nonnull_result_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result	Simplest conditional pattern (False)
direct_optional_null_result_nojs	simplest conditional pattern (true), no javascript
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.

ID	Description
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
docker_output_dir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output

ID	Description
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
exprtool_file_literal	Test file literal output created by ExpressionTool
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null	pickValue: first_non_null second item is non null
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline

ID	Description
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'

ID	Description
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval.ExitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories

ID	Description
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority

ID	Description
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax

ID	Description
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal _with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut _docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_noexp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no source	Test use default value on step input parameter with empty source
step_input_default_value_nu llsource	Test use default value on step input parameter with null source
step_input_default_value_ov erriden	Test default value on step input parameter overridden by provided source
step_input_default_value_ov erriden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_ov erriden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_ov erriden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_ov erriden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_ov erriden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up

ID	Description
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_from_expression_wf	Test expression in time limit in workflow
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)

ID	Description
wf_multiplesources_multipl etypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct _oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossprod uct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_value From	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valuef rom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valuef rom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valuef rom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crosspr oduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_cross product	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput _flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotpr oduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_c rossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_neste d_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow

ID	Description
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after

ID	Description
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

FT3

Failed toil tests on ft3 (local mode) (5 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input

Failed toil tests on ft3 (slurm mode) (7 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
scatter_embedded_subworkf low	Test simple scatter over an embedded subworkflow
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crosspr oduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_cross product	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_c rossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_neste d_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on ft3 (batch mode) (219 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_ default	Test use of \$graph without specifying which process to run
booleanflags_cl_noinputbin ding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line

ID	Description
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.

ID	Description
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nestaddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline

ID	Description
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job.input.secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job.input.subdir.primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
listing_outputBinding.loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings

ID	Description
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
quoting_multiple_backslashes	Test quoting multiple backslashes
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow

ID	Description
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool

ID	Description
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nulsource	Test use default value on step input parameter with null source
step_input_default_value_override	Test default value on step input parameter overridden by provided source
step_input_default_value_override_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_override_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_override_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_override_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_override_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
success_codes	Test successCodes
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
valuefrom_constant_override_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool

ID	Description
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valuefrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valuefrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valuefrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior

ID	Description
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

Failed toil tests on ft3 (all modes union) (220 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement

ID	Description
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.

ID	Description
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles

ID	Description
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input

ID	Description
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions

ID	Description
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.

ID	Description
legal_symlink	symlink to file inside of working directory should be retrieved
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
quoting_multiple_backslashes	Test quoting multiple backslashes
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)

ID	Description
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal

ID	Description
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut _docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
success_codes	Test successCodes
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs

ID	Description
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_no_js	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input

ID	Description
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_nomultiple	Test that no MultipleInputFeatureRequirement is necessary when workflow step source is a single-item list
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after

ID	Description
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

MN5

Failed toil tests on mn5 (local mode) (4 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input

Failed toil tests on mn5 (slurm mode) (45 unique failures)

ID	Description
FAIL_NAME	symlink to file inside of working directory should be retrieved

ID	Description
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
directory_input_docker	Test directory input in Docker
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
hints_unknown_ignored	Test unknown hints are ignored.
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_file_literal	Test file literal as input
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
metadata	Test metadata
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
packed_import_schema	SchemaDefRequirement with \$import, and packed
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
wf_compound_doc	Test compound workflow document
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_simple	Test simple workflow

ID	Description
writable_stagedfiles	Test writable staged files.

Failed toil tests on mn5 (batch mode) (226 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
booleanflags.cl.noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl.basic_generation	General test of command line generation
cl.empty_array_input	Test that empty array input does not add anything to command line
cl.gen_arrayofarrays	Test command line generation of array-of-arrays
cl.optional_bindings_provided	Test command line with optional input (provided)
cl.optional_inputs_missing	Test command line with optional input (missing)
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.

ID	Description
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
conditional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
conditional_scatter_on_nonscattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fail	Non-boolean values from "when" should fail
conditionals_non_boolean_fail_nojs	Non-boolean values from "when" should fail; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores

ID	Description
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result_nojs	simplest conditional pattern (true), no javascript
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript

ID	Description
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)

ID	Description
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.

ID	Description
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript

ID	Description
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_missing	Test checking when secondaryFiles are missing
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename

ID	Description
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step

ID	Description
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_from_expression_wf	Test expression in time limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_wf_step	Test valueFrom on workflow step.
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow

ID	Description
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_parseInt	Test two step workflow with imported tools
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

Failed toil tests on mn5 (all modes union) (228 unique failures)

ID	Description
FAIL_NAME	symlink to file inside of working directory should be retrieved
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter

ID	Description
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
capture_files_and_dirs	Test that both files and directories are captured by glob evaluation when type is [Directory, File]
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_output_path	Confirm that colons are tolerated in output directory names
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nons cattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nons cattered_true_nojs	Simple scatter: conditional on a non scattered variable (True); no javascript
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_ nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_sc atter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_sc atter_nojs	nested cross product scatter with condition on one dimension; no javascript
conditionals_non_boolean_fa il	Non-boolean values from "when" should fail
conditionals_non_boolean_fa il_nojs	Non-boolean values from "when" should fail; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_ expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_ static	Test conflicting requirements in input document via EnvVarRequirement
cwloutput_nolimit	Test that loading from cwl.output.json isn't limited to 64k
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_res ult_nojs	simplest conditional pattern (false), no javascript
direct_optional_null_result_ nojs	simplest conditional pattern (true), no javascript
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal _file_in_subdir_nostdin	Test non-stdin reference to literal File via a nested Directory literal

ID	Description
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement

ID	Description
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_all_null_nojs	pickValue: first_non_null needs at least one non null; no javascript
first_non_null_first_non_null_nojs	pickValue: first_non_null first item is non null; no javascript
first_non_null_second_non_null_nojs	pickValue: first_non_null second item is non null; no javascript
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
format_checking_subclass	Test format checking against ontology using subclassOf.
glob_outside_outputs_fails	Test fail trying to glob outside output directory
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
illegal_symlink	symlink to file outside of working directory should NOT be retrieved
initial_work_dir_for_array_dirs	Test array of directories InitialWorkDirRequirement
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_output_glob	Test full path glob output of InitialWorkDir

ID	Description
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nestaddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initworkdir_expreng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-container-entryname1	Test input mount locations when container required
iwd-container-entryname2	Test input mount locations when no container (should fail)
iwd-container-entryname3	Test input mount locations when container is a hint (should fail)
iwd-container-entryname4	Must fail if entryname starts with ../
iwd-fileobjs1	Test File and Directory object in listing
iwd-fileobjs2	Test File and Directory object in listing
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump1-nl	Test dump object to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2-nl	Test array to JSON in InitialWorkDir file contents, with trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json

ID	Description
iwd-passthrough3	Test InitialWorkDir file object is passed through
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwd-subdir	Test emitting a subdirectory from initial workdir
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
json_output_location_relative	Test support for reading cwl.output.json where 'location' is relative reference to output dir.
json_output_path_relative	Test support for reading cwl.output.json where 'path' is relative path in output dir.
legal_symlink	symlink to file inside of working directory should be retrieved
length_for_non_array	Test parameter reference that refers to length of non-array input
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
metadata	Test metadata
mixed_version_v10_wf	test v1.0 workflow document that runs other versions
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
networkaccess_disabled	Test networkaccess is disabled by default

ID	Description
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
params_broken_null	Test parameter reference that refers to null.something
pass_through_required_fail_nojs	pickValue: the_only_non_null will fail due to multiple non nulls; no javascript
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
quoting_multiple_backslashes	Test quoting multiple backslashes
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_format	Test format on output record fields
record_outputeval	Use of outputEval on a record itself, not the fields of the record
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
record_with_default	Confirm that records with defaults are accepted
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.

ID	Description
scatter_embedded_subworkf low	Test simple scatter over an embedded subworkflow
scatter_multi_input_embed ded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditi onal_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enu m_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_output_re cords	Test secondaryFiles on output record fields
secondary_files_missing	Test checking when secondaryFiles are missing
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_basename	Test that array of input files can be staged to directory with basename
stage_file_array_entryname_ overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
staging-basename	Use of expression tool to change basename of file, then correctly staging the file using the new name.
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal _with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal _with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shorcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection

ID	Description
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
storage_float	Test float value for ram/tmpdir/outdir Min(Max) is rounded up
the_only_non_null_multi_true_nojs	pickValue: the_only_non_null will fail with two active nodes; no javascript
the_only_non_null_single_true_nojs	pickValue: the_only_non_null will pass for only one active node; no javascript
timelimit_from_expression_wf	Test expression in time limit in workflow
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_wf_step	Test valueFrom on workflow step.

ID	Description
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_parseInt	Test two step workflow with imported tools
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow

ID	Description
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
writable_stagedfiles	Test writable staged files.

LUMI

Failed toil tests on lumi (local mode) (7 unique failures)

ID	Description
exprtool_file_literal	Test file literal output created by ExpressionTool
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on lumi (slurm mode) (5 unique failures)

ID	Description
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark

ID	Description
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input

Failed toil tests on lumi (batch mode) (268 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation

ID	Description
cl.empty_array_input	Test that empty array input does not add anything to command line
cl.gen_arrayofarrays	Test command line generation of array-of-arrays
cl.optional_bindings_provided	Test command line with optional input (provided)
cl.optional_inputs_missing	Test command line with optional input (missing)
clt.any_input_with_file_provided	Test Any parameter with file input to a tool
clt.any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt.any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt.any_input_with_record_provided	Test Any parameter with record input to a tool
clt.any_input_with_string_provided	Test Any parameter with string input to a tool
clt.file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt.file_size_property_with_multi_file	Test use of size in expressions for a few files
clt.optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt.optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt.optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt.optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case

ID	Description
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nons cattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nons cattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nons cattered_true	Simple scatter: conditional on a non scattered variable (True)
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_ nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_sc atter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_sc atter_nojs	nested cross product scatter with condition on one dimension; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_ expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_ static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_war ning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_res ult	Simplest conditional pattern (True)
direct_optional_null_result	Simplest conditional pattern (False)
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal _file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output

ID	Description
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrpoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them

ID	Description
expr.reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.
first_non_null_second_non_null	pickValue: first_non_null second item is non null
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions

ID	Description
inlinejs_req-expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_rekurs_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.

ID	Description
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexp	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval.ExitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories

ID	Description
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_for_mat	Test format on output record fields
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow_low	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow

ID	Description
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def-anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection

ID	Description
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source
step_input_default_value_overridden_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_overridden_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_overridden_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_overridden_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_overridden_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs

ID	Description
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multiple_types_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method

ID	Description
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test when step source is a single-item list and there is no linkMerge, then it not wrapped in a list, and that MultipleInputFeatureRequirement is not required.
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior

ID	Description
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.

Failed toil tests on lumi (all modes union) (270 unique failures)

ID	Description
FAIL_NAME	Test null and array input in InitialWorkDirRequirement
all_non_null_all_null	pickValue: all_non_null will produce a list, even if empty
all_non_null_all_null_nojs	pickValue: all_non_null will produce a list, even if empty; no javascript
all_non_null_multi_non_null	pickValue: all_non_null will produce a list
all_non_null_multi_non_null_nojs	pickValue: all_non_null will produce a list; no javascript
all_non_null_one_non_null	pickValue: all_non_null will produce a list; even if single item list
all_non_null_one_non_null_nojs	pickValue: all_non_null will produce a list; even if single item list; no javascript
anonymous_enum_in_array	Test an anonymous enum inside an array inside a record
any_input_param	Test Any type input parameter
any_input_param_graph_no_default	Test use of \$graph without specifying which process to run
any_input_param_graph_no_default_hashmain	Test use of \$graph without specifying which process to run, hash-prefixed "main"
booleanflags_cl_noinputbinding	Test that boolean flags do not appear on command line if inputBinding is empty and not null
cat_synthetic_file	Test the generation of a synthetic file with inline contents.
cl_basic_generation	General test of command line generation
cl_empty_array_input	Test that empty array input does not add anything to command line
cl_gen_arrayofarrays	Test command line generation of array-of-arrays
cl_optional_bindings_provided	Test command line with optional input (provided)
cl_optional_inputs_missing	Test command line with optional input (missing)
clt_any_input_with_file_provided	Test Any parameter with file input to a tool
clt_any_input_with_integer_provided	Test Any parameter with integer input to a tool
clt_any_input_with_mixed_array_provided	Test Any parameter with array input to a tool
clt_any_input_with_record_provided	Test Any parameter with record input to a tool
clt_any_input_with_string_provided	Test Any parameter with string input to a tool

ID	Description
clt_file_size_property_with_empty_file	Test use of size in expressions for an empty file
clt_file_size_property_with_multi_file	Test use of size in expressions for a few files
clt_optional_union_input_file_or_files_with_array_of_one_file_provided	Test input union type or File or File array to a tool with one file in array specified.
clt_optional_union_input_file_or_files_with_many_files_provided	Test input union type or File or File array to a tool with a few files in array specified.
clt_optional_union_input_file_or_files_with_nothing_provided	Test input union type or File or File array to a tool with null specified.
clt_optional_union_input_file_or_files_with_single_file_provided	Test input union type or File or File array to a tool with one file specified.
colon_in_paths	Confirm that colons are tolerated in input paths, string values, stdout shortcut, and output file & directory names
command_input_file_expression	Confirm CommandInputParameter expression can receive a File object
command_output_file_expression	Confirm CommandOutputParameter expression can receive a File object
cond-with-defaults-1	Default inputs, choose step to run based on what was provided, first case
cond-with-defaults-2	Default inputs, choose step to run based on what was provided, second case
condifional_scatter_on_nonscattered_false	Simple scatter: conditional on a non scattered variable (False)
condifional_scatter_on_nonscattered_false_nojs	Simple scatter: conditional on a non scattered variable (False); no javascript
condifional_scatter_on_nonscattered_true	Simple scatter: conditional on a non scattered variable (True)
conditionals_multi_scatter	Scatter two steps, flatten result + pickValue
conditionals_multi_scatter_nojs	Scatter two steps, flatten result + pickValue; no javascript
conditionals_nested_cross_scatter	nested cross product scatter with condition on one dimension
conditionals_nested_cross_scatter_nojs	nested cross product scatter with condition on one dimension; no javascript
continuation	Line continuations in bash scripts should behave correctly
continuation_expression	Line continuations in bash scripts should always behave correctly

ID	Description
cores_float	Test float value for coresMin(Max) is rounded up in runtime.cores
cwl_requirements_addition	Test requirements in input document via EnvVarRequirement
cwl_requirements_override_expression	Test conflicting requirements in input document via EnvVarRequirement and expression
cwl_requirements_override_static	Test conflicting requirements in input document via EnvVarRequirement
default_path_notfound_warning	Test warning instead of error when default path is not found
default_with_falsey_value	Confirms that "false"-like (but not 'null') values override any default
direct_optional_nonnull_result	Simplest conditional pattern (True)
direct_optional_null_result	Simplest conditional pattern (False)
direct_required	Conditional using intermediate WorkflowStep.in input
direct_required_nojs	Conditional using intermediate WorkflowStep.in input; no javascript
directory_input_docker	Test directory input in Docker
directory_input_param_ref	Test directory input with parameter reference
directory_literal_with_literal_file_nostdin	Test non-stdin reference to literal File via a Directory literal
directory_output	Test directory output
directory_secondaryfiles	Test directories in secondaryFiles
docker_entrypoint	Test Docker ENTRYPOINT usage
docker_json_output_location	Test support for reading cwl.output.json when running in a Docker container and just 'location' is provided.
docker_json_output_path	Test support for reading cwl.output.json when running in a Docker container and just 'path' is provided.
dockeroutputdir	Test dockerOutputDirectory
dynamic_initial_workdir	Test dynamic initial work dir
dynamic_resreq_filesizes	Test dynamic resource reqs referencing the size of Files inside a Directory
dynamic_resreq_inputs	Test dynamic resource reqs referencing inputs
dynamic_resreq_wf	Test simple workflow with a dynamic resource requirement

ID	Description
dynamic_resreq_wf_optional_file_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The CommandLineTool input has a default value (a local file) and the workflow nor the workflow step does not provide any value for this input.
dynamic_resreq_wf_optional_file_step_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow step provides a default value (a local file) for this input and the workflow itself does not provide any value for this input.
dynamic_resreq_wf_optional_file_wf_default	Within a workflow, test accessing the size attribute of an optional input File as part of a CommandLineTool's ResourceRequirement calculation. The workflow itself provides a default value (a local file) for this input.
embedded_subworkflow	Test embedded subworkflow
env_home_tmpdir	Test \$HOME and \$TMPDIR are set correctly
env_home_tmpdir_docker	Test \$HOME and \$TMPDIR are set correctly in Docker
env_home_tmpdir_docker_no_return_code	Test \$HOME and \$TMPDIR are set correctly in Docker without using return code
envvar_req	Test EnvVarRequirement
escaping_expression_no_extra_quotes	Strings returned from JS expressions should not have extra quotes around them
expr_reference_self_noinput	Test that expression engine does not fail to evaluate reference to self with unprovided input
expression_any_null	Test explicitly passing null to Any type inputs with default values.
expression_outputEval	Test outputEval to transform output
expressionlib_tool_wf_override	Test that expressionLib requirement of individual tool step overrides expressionLib of workflow.
exprtool_file_literal	Test file literal output created by ExpressionTool
fileliteral_input_docker	Test file literal as input without Docker
filename_with_hash_mark	Test for correct handling of URL-quoting in filename to refer to filename containing a hash mark
filesarray_secondaryfiles	Test required, optional and null secondaryFiles on array of files.

ID	Description
first_non_null_second_non_null	pickValue: first_non_null second item is non null
format_checking	Test simple format checking.
format_checking_equivalent_class	Test format checking against ontology using equivalentClass.
hints_import	Test hints with \$import
hints_unknown_ignored	Test unknown hints are ignored.
initial_work_dir_for_null_and_arrays	Test null and array input in InitialWorkDirRequirement
initial_workdir_empty_writable	Test empty writable dir with InitialWorkDirRequirement
initial_workdir_empty_writable_docker	Test empty writable dir with InitialWorkDirRequirement inside Docker
initial_workdir_expr	Test expression in InitialWorkDir listing
initial_workdir_output	Test output of InitialWorkDir
initial_workdir_secondary_files_expr	Test InitialWorkDirRequirement linking input files and capturing secondaryFiles on input and output. Also tests the use of a variety of parameter references and expressions in the secondaryFiles field.
initial_workdir_trailingnl	Test if trailing newline is present in file entry in InitialWorkDir
initialworkdir_nesteddir	Test InitialWorkDirRequirement with a nested directory structure from another step
initialworkpath_output	Test that file path in \$(inputs) for initialworkdir is in \$(outdir).
initworkdir_expr_eng_requirements	Test InitialWorkDirRequirement ExpressionEngineRequirement.engineConfig feature
inline_expressions	Test inline expressions
inlinejs_req_expressions	Test InlineJavascriptRequirement with multiple expressions in the same tool
inputBinding_position_expr	Test for expression in the InputBinding.position field; also test using emoji in CWL document and tool output
input_dir_inputbinding	Test directory input with inputBinding
input_dir_recurse_copy_writable	Test if a writable input directory is recursively copied and writable
input_file_literal	Test file literal as input
iwd-jsondump1	Test dump object to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump2	Test array to JSON in InitialWorkDir file contents, no trailing newline

ID	Description
iwd-jsondump3	Test number to JSON in InitialWorkDir file contents, no trailing newline
iwd-jsondump3-nl	Test number to JSON in InitialWorkDir file contents, with trailing newline
iwd-nolimit	Test that InitialWorkDir contents can be bigger than 64k
iwd-passthrough1	Test InitialWorkDir file passthrough
iwd-passthrough2	Test InitialWorkDir file object is serialized to json
iwd-passthrough4	Test InitialWorkDir file object is passed through
iwdr_dir_literal_real_file	Test directory literal containing a real file
job_input_secondary_subdirs	Test specifying secondaryFiles in subdirectories of the job input document.
job_input_subdir_primary_and_secondary_subdirs	Test specifying secondaryFiles in same subdirectory of the job input as the primary input file.
listing_default_none	Test that default behavior is 'no_listing' if not specified
listing_loadListing_deep	Test that 'listing' is present in top directory object and in subdirectory objects when input parameter loadListing is 'deep_listing'
listing_loadListing_none	Test that 'listing' is not present when loadListing on input parameter is 'no_listing'
listing_loadListing_shallow	Test that 'listing' is present in top directory object but not subdirectory object when loadListing on input parameter loadListing is 'shallow_listing'
listing_outputBinding_loadListing	Test that loadListing works when used on outputBinding.
listing_requirement_deep	Test that 'listing' is present in top directory object and in subdirectory objects when LoadListingRequirement is 'deep_listing'
listing_requirement_none	Test that 'listing' is not present when LoadListingRequirement is 'no_listing'
listing_requirement_shallow	Test that 'listing' is present in top directory object but not subdirectory object when LoadListingRequirement is 'shallow_listing'
metadata	Test metadata
mixed_version_v11_wf	test v1.1 workflow document that runs other versions
mixed_version_v12_wf	test v1.2 workflow document that runs other versions

ID	Description
modify_directory_content	inplace update has side effect on directory content
modify_file_content	inplace update has side effect on file content
multiple-input-feature-requirement	MultipleInputFeatureRequirement on workflow outputs.
multiple_glob_expr_list	Test support for returning multiple glob patterns from expression
nameroot_nameext_generated	Test that nameroot and nameext are generated from basename at execution time by the runner
nameroot_nameext_stdout_expr	Test nameroot/nameext expression in arguments, stdout
nested_cl_bindings	Test nested command line bindings
nested_prefixes_arrays	Test nested prefixes with arrays
nested_types	demonstrate usage of nested types
nested_workflow	Test nested workflow
nested_workflow_noexpr	Test nested workflow, without ExpressionTool
networkaccess	Test networkaccess enabled
no_inputs_commandlinetool	Test CommandLineTool without inputs
no_inputs_workflow	Test Workflow without inputs
no_outputs_commandlinetool	Test CommandLineTool without outputs
no_outputs_workflow	Test Workflow without outputs
null_missing_params	Test that missing parameters are null (not undefined) in expression
optional_numerical_output_returns_0_not_null	Test that optional number output is returned as 0, not null
outputEval_exitCode	Can access exit code in outputEval
output_secondaryfile_optional	Test optional output file and optional secondaryFile on output.
outputbinding_glob_directory	Test that OutputBinding.glob accepts Directories
outputbinding_glob_sorted	Test that OutputBinding.glob is sorted as specified by POSIX
packed_import_schema	SchemaDefRequirement with \$import, and packed
param_evaluation_expr	Test parameter evaluation, with support for JS expressions
param_evaluation_noexpr	Test parameter evaluation, no support for JS expressions
param_notnull_expr	Test that provided parameter is not null in expression
pass_through_required_false_when	Pass through pattern with pickValue: first_non_null; 'when' is false

ID	Description
pass_through_required_false_when_nojs	Pass through pattern with pickValue: first_non_null; 'when' is false'; no javascript
pass_through_required_the_only_non_null	pickvalue: the_only_non_null will pass, only for false condition
pass_through_required_the_only_non_null_nojs	pickvalue: the_only_non_null will pass, only for false condition; no javascript
pass_through_required_true_when	pass through pattern with pickvalue: first_non_null; 'when' is true
pass_through_required_true_when_nojs	pass through pattern with pickvalue: first_non_null; 'when' is true'; no javascript
record_order_with_input_bindings	Test sorting arguments at each level (inputBinding for all levels)
record_output_binding	Test record type output binding.
record_output_file_entry_for_mat	Test format on output record fields
record_outputeval_nojs	Use of outputEval on a record itself, not the fields of the record (without javascript)
rename	Test InitialWorkDirRequirement with expression in filename.
requirement_override_hints	Test requirements override hints
requirement_priority	Test requirement priority
requirement_workflow_steps	Test requirements on workflow steps
resreq_step_overrides_wf	Test that ResourceRequirement on a step level redefines requirement on the workflow level
runtime-outdir	Use of \$(runtime.outdir) for outputBinding glob.
scatter_embedded_subworkflow	Test simple scatter over an embedded subworkflow
scatter_multi_input_embedded_subworkflow	Test simple multiple input scatter over an embedded subworkflow
scatter_on_scattered_conditional	Simple scatter: Add conditional variable to scatter
scatter_on_scattered_conditional_nojs	Simple scatter: Add conditional variable to scatter; no javascript
schema-def_anonymous_enum_in_array	Test an anonymous enum inside an array inside a record, SchemaDefRequirement
schemadef_req_tool_param	Test SchemaDefRequirement definition used in tool parameter
schemadef_req_wf_param	Test SchemaDefRequirement definition used in workflow parameter
secondary_files_in_named_records	Test secondaryFiles on SchemaDefRequirement record fields

ID	Description
secondary_files_in_output_records	Test secondaryFiles on output record fields
secondary_files_in_unnamed_records	Test secondaryFiles on anonymous record fields
secondary_files_missing	Test checking when secondaryFiles are missing
secondary_files_workflow_propagation	Test secondaryFiles propagation through workflow
shelldir_notinterpreted	Test that shell directives are not interpreted.
shelldir_quoted	Test that shell directives are quoted.
stage_file_array	Test that array of input files can be staged to directory with entryname
stage_file_array_entryname_overrides	Test that if array of input files are staged to directory with basename and entryname, entryname overrides
stderr_redirect	Test command line with stderr redirection
stderr_redirect_mediumcut	Test command line with stderr redirection, named brief syntax
stderr_redirect_shortcut	Test command line with stderr redirection, brief syntax
stdin_from_directory_literal_with_literal_file	Pipe to stdin from literal File via a Directory literal
stdin_from_directory_literal_with_local_file	Pipe to stdin from user provided local File via a Directory literal
stdin_shortcut	Test command execution in with stdin and stdout redirection using stdin shortcut
stdinout_redirect	Test command execution in with stdin and stdout redirection
stdinout_redirect_docker	Test command execution in Docker with stdin and stdout redirection
stdout_redirect_docker	Test command execution in Docker with stdout redirection
stdout_redirect_mediumcut_docker	Test command execution in Docker with mediumcut stdout redirection
stdout_redirect_shortcut_docker	Test command execution in Docker with shortcut stdout redirection
step_input_default_value	Test default value on step input parameter
step_input_default_value_no_exp	Test default value on step input parameter, no ExpressionTool
step_input_default_value_no_source	Test use default value on step input parameter with empty source
step_input_default_value_nullsource	Test use default value on step input parameter with null source
step_input_default_value_overridden	Test default value on step input parameter overridden by provided source

ID	Description
step_input_default_value_override_2nd_step	Test default value on step input parameter overridden by provided source. With passthrough first step
step_input_default_value_override_2nd_step_noexp	Test default value on step input parameter overridden by provided source. With passthrough first step and no ExpressionTool
step_input_default_value_override_2nd_step_null	Test default value on step input parameter overridden by provided source. With null producing first step
step_input_default_value_override_2nd_step_null_noexp	Test default value on step input parameter overridden by provided source. With null producing first step and no ExpressionTool
step_input_default_value_override_noexp	Test default value on step input parameter overridden by provided source, no ExpressionTool
success_codes	Test successCodes
the_only_non_null_multi_true	pickValue: the_only_non_null will fail with two active nodes
the_only_non_null_single_true	pickValue: the_only_non_null will pass for only one active node
timelimit_basic	Test that job fails when exceeding time limit
timelimit_basic_wf	Test that tool in workflow fails when exceeding time limit
timelimit_invalid	Test invalid time limit value
timelimit_invalid_wf	Test that workflow level time limit is not applied to workflow execution time
timelimit_zero_unlimited	Test zero timelimit means no limit
timelimit_zero_unlimited_wf	Test zero timelimit means no limit in workflow
tmpdir_is_not_outdir	Test that runtime.tmpdir is not runtime.outdir
user_defined_length_in_parameter_reference	Test 'length' in parameter reference where it does not refer to length of an array input
valuefrom_constant_overrides_inputs	Test valueFrom with constant value overriding provided array inputs
valuefrom_ignored_null	Test that valueFrom is ignored when the parameter is null
valuefrom_secondexpr_ignored	Test that second expression in concatenated valueFrom is not ignored
valuefrom_wf_step	Test valueFrom on workflow step.
valuefrom_wf_step_multiple	Test valueFrom on workflow step with multiple sources
valuefrom_wf_step_other	Test valueFrom on workflow step referencing other inputs

ID	Description
very_big_and_very_floats_nojs	Confirm that very big and very small numbers are represented using decimals, not scientific notation
wf_compound_doc	Test compound workflow document
wf_default_tool_default	Test that workflow defaults override tool defaults
wf_input_default_missing	Test workflow with default value for input parameter (missing)
wf_input_default_provided	Test workflow with default value for input parameter (provided)
wf_multiplesources_multipletypes_noexp	Test step input with multiple sources with multiple types, without ExpressionTool
wf_scatter_flat_crossproduct_oneempty	Test workflow scatter with two scatter parameters, one of which is empty and flat_crossproduct join method
wf_scatter_nested_crossproduct_firstempty	Test workflow scatter with two scatter parameters and nested_crossproduct join method with first list empty
wf_scatter_oneparam_valueFrom	Test valueFrom eval on scattered input parameter
wf_scatter_oneparam_valueFrom	Test workflow scatter with single scatter parameter and two valueFrom on step input (first and current el)
wf_scatter_oneparam_valueFrom_inputs	Test workflow scatter with single scatter parameter and two valueFrom using \$inputs (first and current el)
wf_scatter_oneparam_valueFrom_twice_current_el	Test workflow scatter with single scatter parameter and two valueFrom on step input (current el twice)
wf_scatter_single_param	Test workflow scatter with single scatter parameter
wf_scatter_two_dotproduct	Test workflow scatter with two scatter parameters and dotproduct join method
wf_scatter_two_flat_crossproduct	Test workflow scatter with two scatter parameters and flat_crossproduct join method
wf_scatter_two_nested_crossproduct	Test workflow scatter with two scatter parameters and nested_crossproduct join method
wf_scatter_twopar_oneinput_flattenedmerge	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior. Workflow inputs are set as list

ID	Description
wf_scatter_twoparam_dotproduct_valuefrom	Test workflow scatter with two scatter parameters and dotproduct join method and valueFrom on step input
wf_scatter_twoparam_flat_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and flat_crossproduct join method and valueFrom on step input
wf_scatter_twoparam_nested_crossproduct_valuefrom	Test workflow scatter with two scatter parameters and nested_crossproduct join method and valueFrom on step input
wf_simple	Test simple workflow
wf_step_access_undeclared_param	Test that parameters that don't appear in the 'run' process inputs are not present in the input object used to run the tool.
wf_step_connect_undeclared_param	Test that it is not an error to connect a parameter to a workflow step, even if the parameter doesn't appear in the 'run' process inputs.
wf_two_inputfiles_namecollision	Test workflow two input files with same name.
wf_wc_expressiontool	Test two step workflow with inline tools
wf_wc_nomultiple	Test when step source is a single-item list and there is no linkMerge, then it not wrapped in a list, and that MultipleInputFeatureRequirement is not required.
wf_wc_nomultiple_merge_nested	Test when step source is a single-item list and linkMerge is listed, then it is wrapped in a list.
wf_wc_parseInt	Test two step workflow with imported tools
wf_wc_scatter	Test single step workflow with Scatter step
wf_wc_scatter_multiple_flattened	Test single step workflow with Scatter step and two data links connected to same input, flattened merge behavior
wf_wc_scatter_multiple_merge	Test single step workflow with Scatter step and two data links connected to same input, default merge behavior
wf_wc_scatter_multiple_nested	Test single step workflow with Scatter step and two data links connected to same input, nested merge behavior
workflow_any_input_with_file_provided	Test Any parameter with file input to a workflow
workflow_any_input_with_integer_provided	Test Any parameter with integer input to a workflow
workflow_any_input_with_mixed_array_provided	Test Any parameter with array input to a workflow

ID	Description
workflow_any_input_with_record_provided	Test Any parameter with record input to a tool
workflow_any_input_with_string_provided	Test Any parameter with string input to a workflow
workflow_embedded_subworkflow_embedded_subsubworkflow	Test twice nested subworkflow
workflow_embedded_subworkflow_with_subsubworkflow_and_tool	Test subworkflow of mixed depth with tool after
workflow_embedded_subworkflow_with_tool_and_subsubworkflow	Test subworkflow of mixed depth with tool first
workflow_file_array_output	Test output arrays in a workflow (with Files).
workflow_file_input_default_specified	Test File input with default specified to workflow
workflow_file_input_default_unspecified	Test File input with default unspecified to workflow
workflow_integer_input_optional_unspecified	Test optional integer workflow inputs (unspecified)
workflow_records_inputs_and_outputs	Test record type inputs to and outputs from workflows.
workflowstep_valuefrom_file_basename	Test valueFrom on workflow step using basename.
workflowstep_valuefrom_string	Test valueFrom on workflow step from literal (string).
writable_stagedfiles	Test writable staged files.